

N-EPI® 1mg/ml Concentrate for Solution for Infusion

Norepinephrine Bitartrate 1 mg

Product Description

Clear solution, colourless to pale yellow colour and negative precipitate.

After reconstitution with 5% Glucose solution, and with 0.9% Sodium chloride solution and 5% Dextrose solution, the solutions of product remain clear, colourless and negative of precipitate until 24 hours.

Composition

Each ml Concentrate for solution for infusion contains Norepinephrine Bitartrate equivalent to 1mg of Noradrenaline.

Pharmacodynamics

The vascular effects in the doses normally used clinically result from the simultaneous stimulation of alpha and beta adrenergic receptors in the heart and vascular system. Except in the heart, its action is predominantly on the alpha receptors. This results in an increase in the force (and in the absence of vagal inhibition, in the rate) of myocardial contraction. Peripheral resistance increases and diastolic and systolic pressures are raised.

The increase in blood pressure may cause a reflex decrease in heart rate. Vasoconstriction may result in decreased blood flow in kidneys, liver, skin and smooth muscles. Local vasoconstriction may cause haemostasis and/or necrosis. The effect on blood pressure disappears 1-2 minutes after stopping the infusion.

Pharmacokinetics

Up to 16% of an intravenous dose is excreted unchanged in the urine with methylated and deaminated metabolites in free and conjugated forms.

Indication

For blood pressure control in certain acute hypotensive states (e.g., pheochromocytectomy, sympathectomy, poliomyelitis, spinal anaesthesia, myocardial infarction, septicemia, blood transfusion, and drug reactions).

As an adjunct in the treatment of cardiac arrest and profound hypotension.

Recommended Dosage

Norepinephrine Bitartrate Injection is a concentrated, potent drug which must be diluted in dextrose-containing solutions prior to infusion. An infusion of N-EPI should be given into a large vein (see Warnings and Precautions).

Restoration of Blood Pressure in the Acute Hypotensive States: Blood volume depletion should always be corrected as fully as possible before any vasopressor is administered. When, as an emergency measure, intraaortic pressures must be maintained to prevent cerebral or coronary artery ischemia, N-EPI can be administered before and concurrently with blood volume replacement.

Diluent: N-EPI should be diluted in 5% glucose injection or 5% dextrose and sodium chloride injections. These dextrose containing fluids are protection against significant loss of potency due to oxidation. **Administration in saline solution alone is not recommended.** Whole blood or plasma, if indicated to increase blood volume, should be administered separately (for example, by use of a Y-tube and individual containers if given simultaneously).

Average Dosage: Add the content of the vial (4 mg/4 mL) of N-EPI to 1,000 mL of a 5 percent dextrose containing solution. Each mL of this dilution contains 4 mcg of the base of N-EPI.

Give this solution by intravenous infusion. Insert a plastic intravenous catheter through a suitable bore needle well advanced centrally into the vein and securely fixed with adhesive tape, avoiding, if possible, a catheter tie-in technique as this promotes stasis. An intravenous drip chamber or other suitable metering device is essential to permit an accurate estimation of the rate of flow in drops per minute. After observing the response to an initial dose of 2 mL to 3 mL (from 8 mcg to 12 mcg of base) per minute, adjust the rate of flow to establish and maintain a low normal blood pressure (usually 80 mm Hg to 100 mm Hg systolic) sufficient to maintain the circulation to vital organs. In previously hypertensive patients, it is recommended that the blood pressure should be raised no higher than 40 mm Hg below the pre-existing systolic pressure. The average maintenance dose ranges from 0.5 mL to 1 mL per min (from 2 mcg to 4 mcg of base).

High Dosage: Great individual variation occurs in the dose required to attain and maintain adequate blood pressure. In all cases, the dosage of N-EPI should be titrated according to the response of the patient. Occasionally much larger or even enormous daily doses (as high as 68 mg base or 17 vials) may be necessary if the patient remains hypotensive, but occult blood volume depletion should always be suspected and corrected when present. Central venous pressure monitoring is usually helpful in detecting and treating this situation.

Fluid Intake: The degree of dilution depends on clinical fluid volume requirements. If large volumes of fluid (dextrose) are needed at a flow rate that would involve an excessive dose of the pressor agent per unit of time, a solution more dilute than 4 mcg per mL should be used. On the other hand, when large volumes of fluid are clinically undesirable, a concentration greater than 4 mcg per mL may be necessary.

Duration of Therapy: The infusion should be continued until adequate blood pressure and tissue perfusion are maintained without therapy. Infusions of N-EPI should be reduced gradually, avoiding abrupt withdrawal. In some of the reported cases of vascular collapse due to acute myocardial infarction, treatment was required for up to six days.

Adjunctive Treatment in Cardiac Arrest: Infusions of N-EPI are usually administered intravenously during cardiac resuscitation to restore and maintain adequate blood pressure after an effective heartbeat and ventilation have been established by other means. [N-EPI's powerful beta-adrenergic stimulating action is also thought to increase the strength and effectiveness of systolic contractions once they occur].

Average Dosage: To maintain systemic blood pressure during the management of cardiac arrest, N-EPI is used in the same manner as described under Restoration of Blood Pressure in Acute Hypotensive States.

Parenteral drug products should be inspected visually for particulate matter and discolouration prior to use, whenever solution and container permit.

Do not use the solution if its colour is pinkish or darker than slightly yellow or if it contains a precipitate. Avoid contact with iron salts, alkali or oxidizing agents.

Route of Administration

Intravenous (IV).

Contraindications

N-EPI should not be given to patients who are hypersensitivity to noradrenaline tartrate or to any of the excipients (Sodium chloride, Sodium Metabisulfite, Citric acid monohydrate, and Sodium citrate dihydrate).

N-EPI should not be given to patients who are hypotensive from blood volume deficits except as an emergency measure to maintain coronary and cerebral artery perfusion until blood volume replacement therapy can be completed. If N-EPI is continuously administered to maintain blood pressure in the absence of blood volume replacement, the following may occur: Severe peripheral and visceral vasoconstriction, decreased renal perfusion and urine output, poor systemic blood flow despite "normal" blood pressure, tissue hypoxia and lactate acidosis.

N-EPI should not also be given to patients with mesenteric or peripheral vascular thrombosis (because of the risk of increasing ischemia and extending the area of infarction) unless, in the opinion of the attending physician, the administration of N-EPI is necessary as a life-saving procedure.

Cyclopropane and halothane anaesthetics increase cardiac autonomic irritability and therefore, seem to sensitize the myocardium to the action of intravenously administered epinephrine or norepinephrine. Hence, the use of N-EPI during cyclopropane and halothane anaesthesia is generally considered contraindicated because of the risk of producing ventricular tachycardia or fibrillation.

The same type of cardiac arrhythmias may result from the use of N-EPI in patients with profound hypoxia or hypercarbia.

Warning and Precautions

Warnings

N-EPI should be used with extreme caution in patients receiving monoamine oxidase inhibitors (MAOI) or antidepressants of the triptyline or imipramine types, because severe, prolonged hypertension may result.

N-EPI contains sodium metabisulfite, a sulfite that may cause allergic-type reactions including anaphylactic symptoms and life-threatening or less severe asthmatic episodes in certain susceptible people. The overall prevalence of sulfite sensitivity in the general population is unknown. Sulfite sensitivity is seen more frequently in asthmatic than in nonasthmatic people.

Special Precautions

General: Avoid Hypertension: Because of the potency of N-EPI and because of varying response to pressor substances, the possibility always exists that dangerously high blood pressure may be produced with overdoses of this pressor agent. It is desirable, therefore, to record blood pressure every two minutes from the time administration is started until the desired blood pressure is obtained, then every five minutes if the administration is to be continued.

The rate of flow must be watched constantly, and the patient should never be left unattended while receiving N-EPI. Headache may be a symptom of hypertension due to overdosage.

Site of Infusion: Whenever possible, infusions of N-EPI should be given into a large vein, particularly an antecubital vein because, when administered into this vein, the risk of necrosis of the overlying skin from prolonged vasoconstriction is apparently very slight. Some authors have indicated that the femoral vein is also an acceptable route of administration. A catheter tie-in technique should be avoided, if possible, since the obstruction to blood flow around the tubing may cause stasis and increased local concentration of the drug. Occlusive vascular diseases (for example, atherosclerosis, arteriosclerosis, diabetic endarteritis, Buerger's disease) are more likely to occur in the lower than in the upper extremity.

Therefore, one should avoid the veins of the leg in elderly patients or in those suffering from such disorders. Gangrene has been reported in a lower extremity when infusions of N-EPI were given in an ankle vein.

Extravasation: The infusion site should be checked frequently for free flow. Care should be taken to avoid extravasation of N-EPI into the tissues, as local necrosis might ensue due to the vasoconstrictive action of norepinephrine. Blanching along the course of the infused vein, sometimes without obvious extravasation, has been attributed to vasa vasorum constriction with increased permeability of the vein wall, permitting some leakage.

This also may progress on rare occasions to superficial slough, particularly during infusion into leg veins in elderly patients or in those suffering from obliterative vascular disease. Hence, if blanching occurs, consideration should be given to the advisability of changing the infusion site at intervals to allow the effects of local vasoconstriction to subside.

IMPORTANT: Antidote for Extravasation Ischemia: To prevent sloughing and necrosis in areas in which extravasation has taken place, the area should be infiltrated as soon as possible with 10 mL to 15 mL of saline solution containing 5 mg to 10 mg of Regitine (brand of phentolamine), an adrenergic blocking agent. A syringe with a fine hypodermic needle should be used, with the solution being infiltrated liberally throughout the area, which is easily identified by its cold, hard and pallid appearance. Sympathetic blockade with phentolamine causes immediate and conspicuous local hyperemic changes if the area is infiltrated within 12 hours. Therefore, phentolamine should be given as soon as possible after the extravasation is noted.

Carcinogenesis, Mutagenesis, Impairment of Fertility: Studies have not been performed.

Pediatric Use: Safety and effectiveness in paediatric patients have not been established.

Geriatric Use: Reported clinical experience has not identified differences in responses between the elderly and younger patients. In general, dose selection for an elderly patient should be cautious, usually starting at the low end of the dosing range, reflecting the greater frequency of decreased hepatic, renal, or cardiac function, and of concomitant disease or other drug therapy.

N-EPI should not be administered into the veins in the leg in elderly patients (see Precautions).

Interactions with Other Medicaments

The use of noradrenaline with volatile halogenated anaesthetic agents, monoamine oxidase inhibitors, linezolid, tricyclic antidepressants, adrenergic serotonergic drugs or any other cardiac sensitizing agents is not recommended because severe, prolonged hypertension and possible arrhythmias may result.

Pregnancy and Lactation

Pregnancy

Noradrenaline may impair placental perfusion and induce fetal bradycardia. It may also exert a contractile effect on the pregnant uterus and lead to fetal asphyxia in late pregnancy. These possible risks to the fetus should therefore be weighed against the potential benefit to the mother.

Breastfeeding

No information is available on the use of noradrenaline in lactation.

Side Effects

System Organ Class	Undesirable effect
Psychiatric disorders	Anxiety
Nervous system disorders	Headache
Cardiac disorders	Arrhythmias (when used in conjunction with cardiac sensitizing agents), bradycardia, stress cardiomyopathy
Vascular disorders	Hypertension, peripheral ischaemia including gangrene of the extremities, plasma volume depletion with prolonged use
Respiratory, thoracic and mediastinal disorders	Dyspnoea
General disorders and administration site conditions	Extravasation necrosis at injection site

Symptoms and Treatment of Overdose

Overdosage may result in severe hypertension, reflex bradycardia, marked increased in peripheral resistance and decreased cardiac output. These may be accompanied by a violent headache, photophobia, retrosternal pain, pallor, intense sweating and vomiting. In the event of overdosage, treatment should be withdrawn and appropriate corrective treatment initiated.

Effects on Ability to Drive and Use Machine

None stated.

Instructions for Use

Dilution instructions:

Dilute before use with glucose 5% solution or sodium chloride 9 mg/ml (0.9%) with glucose 5 % solution.

Either add 2 ml concentrate to 48 ml glucose 5% solution (or sodium chloride 9 mg/ml (0.9%) with glucose 5% solution) for administration by syringe pump, or add 20 ml of concentrate to 480 ml glucose 5 % solution (or sodium chloride 9 mg/ml (0.9%) with glucose 5% solution) for administration by drip counter. In both cases the final concentration of the infusion solution is 40 mg/litre noradrenaline base (which is equivalent to 80 mg/litre noradrenaline tartrate). Dilutions other than 40 mg/litre noradrenaline base may also be used (see section Recommended Dose). If dilutions other than 40 mg/litre noradrenaline base are used, check the infusion rate calculation carefully before starting treatment.

The product is compatible with PVC infusion bags.

Incompatibilities

Infusion solutions containing noradrenaline tartrate have been reported to be incompatible with the following substances: alkalis and oxidizing agents, barbiturates, chlorpheniramine, chlorothiazide, nitrofurantoin, novobiocin, phenytoin, sodium bicarbonate, sodium iodide and streptomycin.

Storage Conditions

Store at below 30°C and protect from light.

After dilution: Store at room temperature (25° - 30°C) and cold room (2° - 8° C) for 24 hours after dilution.

Shelf Life

24 months.

After dilution: N-EPI which is reconstituted with 5% Glucose solution and with 0.9% Sodium Chloride and 5% Dextrose solution stored at room temperature (25° - 30°C) and cold room (2° - 8° C) are stable for 24 hours.

Dosage form and Packaging Available

Concentrate for Solution for Injection- Box of 5 ampoules @ 4ml

Registration Number

N-EPI® 1mg/ml Concentrate for Solution for Infusion - MALXXXXXXXXA

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